electricity-demand-sri-lanka

Electricity Demand Intelligence System (EDIS) for Sri Lanka using a trained CNN-LSTM model, served via FastAPI and visualized in a React dashboard.

What This App Is

- A dataset-backed inference dashboard.
- The backend loads a pre-trained model (`.keras`) and scalers (`.pkl`).
- Forecasts are generated from the most recent timestamp available in the bundled CSV.

What This App Is Not (Yet)

- Not a true real-time 2026 national demand monitor.
- Not connected to a live utility/SCADA feed by default.
- Not automatically retraining on new incoming data.

Quick Start

Backend

```
python -m uvicorn api.main:app --reload --port 8000
```

Frontend

```
cd frontend
npm install
npm run dev
```

Configuration

The backend supports environment configuration:

- `MODEL_PATH` (default: `outputs/best_model_new.keras`)
- `FEAT_SCALER_PATH` (default: `outputs/feat_scaler.pkl`)
- `TARGET_SCALER_PATH` (default: `outputs/target_scaler.pkl`)
- `DATA_PATH` (default: `outputs/electricity_demand_srilanka.csv`)
- `ALLOWED_ORIGINS` (comma-separated local origins for CORS)
- `DATA_PROVIDER_MODE` (`dataset_backed_demo` or `live_stream`)
- `DATA_STALE_THRESHOLD_MINUTES` (default: `120`)

The frontend supports:

- `VITE_API_BASE_URL` (default: `/api`)
- `VITE_FORECAST_STEPS` (default: `24`)
- `VITE_FORECAST_PASSES` (default: `3`)
- `VITE_API_TIMEOUT_MS` (default: `60000`)

Health Endpoint

`GET /health` now returns:

- readiness flags (`model`, `data`)
- row count
- app mode (`dataset_backed_demo`)
- live-mode flag (`live_data_enabled`)
- oldest/latest dataset timestamp
- data freshness (`data_age_minutes`, `is_data_stale`)
- feature metadata for debugging deployment mismatches

`GET /data/source` returns provider mode and freshness summary used by the frontend mode badge.

Production Notes

See `docs/PRODUCTION_READY_NOTES.md` for:

- fixes applied in this repository
- why this is currently a dataset-backed demo
- remaining work for true live production readiness